

# CONSCIENTIOUSNESS AND LABOUR MARKET OUTCOMES: A NIDS ANALYSIS

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## Abstract

Abstract to be written here. The abstract should not be too long and should provide the reader with a good understanding what you are writing about. Academic papers are not like novels where you keep the reader in suspense. To be effective in getting others to read your paper, be as open and concise about your findings here as possible. Ideally, upon reading your abstract, the reader should feel he / she must read your paper in entirety.

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## 1. Introduction

## 2. Motivation

Many papers have shown the benefits of cultivating positive personality traits in children for various labour market outcomes at a later stage. A lesser researched question involves the relationship which conscientiousness<sup>1</sup>, specifically, has with labour market outcomes. This measure is unobservable, however, it is hypothesised to affect outcomes such as wages and employment status. Recording this relationship in a linear model shows that the effect decreases to a negligible magnitude after conditioning on relevant control variables. This result encourages the use of alternative means of estimation to avoid the loss of the importance of this variable and to uncover its true effects. In particular, a gap exists in literature of this relationship in a South African context. Much of the current research focuses on developed-country contexts which, as a result, can fail to be externally valid to environments outside of this developmental level. Thus, this paper also aims to close part of this literature gap with the intention of aiding policy formulation around childhood development in South Africa.

## 3. Literature Review

Vast amounts of literature shows that personality is an important estimator for the labour outcomes which an individual achieves. Heckman, Stixrud & Urzua (2006) provides the most comprehensive study of this research question. Using national survey data from the United States, they find that both cognitive and non-cognitive skills explain a multitude of labour outcomes, such as schooling decisions, wages earned, employment status, work experience and choice of occupation. Their approach differs to the literature before it in that they use more advanced empirical methods to address issues of reverse causality which may be found when examining schooling and family influence, and their effects on measured cognitive and non-cognitive skills.

Fletcher (2013) finds in a similar study that when accounting for sibling differences to estimate the effect of personality on employment and wages, that the former has significant effects on the latter for adults. The use of sibling comparisons allowed the authors to control for background family factors which could drive the relationship between personality and labour market outcomes. The results question the robustness of the relationship between conscientiousness and wages earned, a link which had been well-established in prior literature. Novel information was obtained through the finding that extraversion is a significant estimator for labour outcomes. In a paper by Cubel, Nuevo-Chiquero, Sanchez-Pages & Vidal-Fernandez (2016), the mechanism through which personality interacts with labour outcomes was examined. Data was obtained through an experiment ran in Australia. The authors found that productivity was the instrument which is affected by personality, which led to changes in one's labour market outcomes. More specifically

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<sup>1</sup>Relating to the quality of wanting to do one's work well, and measures self-control, diligence and attention to detail, and one of the 'Big Five' personality traits (Oxford University Press, n.d.).

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on conscientiousness, those who scored higher for this measure were found to be more productive and consequently had higher earnings.

Allemand, Kirchberger, Milusheva, Newman, Roberts & Thorne (2023) study the role which conscientiousness has on labour market returns. They performed a randomised control trial in Senegal to determine the effect which the personality measure can have on labour outcomes. Results showed that improving conscientiousness through a training program can play an important role in increasing the amount of time that workers keep their jobs and the earnings received from it. The study contributed to the literature gap by highlighting the effectiveness of such an intervention in a developing-country context, which had not been presented before.

Research done by Prevoo & ter Weel (2013) showed that conscientiousness is also found to be important for socio-economic outcomes. Using data from a British cohort study the authors found a significant and large correlation between early conscientiousness and outcomes in adulthood, such as wages earned, levels of individual savings, life satisfaction, and propensity to engage in drinking or smoking. This paper helped to show the importance of conscientiousness in predicting later-life outcomes in individuals.

While the papers presented above provide important background for the proposed study, the results lack external validity to be implemented in a South African context. Thus, research using data from the country can help close the literature gap which currently exists on the topic in South Africa.

## 4. Methodology

Various empirical techniques were employed in this paper. This section highlights the methods used throughout, following the progression of technical complexity which was highlighted in the motivation. It starts with a description of the data used, before explaining the process of estimation.

### 4.1. Data

Data for this paper was obtained from the National Income Dynamics Study (NIDS). The study took the form of a five-wave longitudinal study which surveyed a nationally representative sample of households and individuals across South Africa over the years 2008 to 2017.

The sample was limited to those individuals between the ages of 14 and 23 in wave 1. This allowed for the calculation of their conscientiousness scores at the same wave, and subsequently for these scores to be regressed on their employment statuses and wages earned in 9 years time at wave 5 of the study. Questions were asked across a variety of themes including demographics, household makeup, education, income, personal asset ownership and debt, as well as health amongst other themes. From these questions, information pertaining to labour market outcomes, conscientiousness proxies, and other important information relevant to the empirical analysis was obtained.

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## 4.2. Analysis

Many methods were used to establish the role which conscientiousness had on labour market outcomes. This section lays out the empirical steps taken to capture this relationship, starting with the construction of the conscientiousness measure.

As no measure of conscientiousness existed in the dataset, four proxy questions from the first wave of the data were used in the assembling of this measure. A principal components analysis was then used to reduce dimensionality of these questions into a single measure<sup>2</sup>. Thereafter, the resulting continuous variable was able to be used as a proxy measure for conscientiousness, with higher values indicating a greater conscientiousness score and lower values meaning lower scores. Initially, the assumption was made that conscientiousness enters the data generating process of wages in a linearly-additive manner. This was shown from the setup in equation 1, where  $c_i$  represents  $i$ 's conscientiousness score,  $\mathbf{X}_i$  represents a vector of controls for individual  $i$  and  $\varepsilon_i$  is their error term.

$$\boxed{\log(Wages_i) = \beta_{0,i} + \alpha c_i + \mathbf{X}'_i \boldsymbol{\delta} + \varepsilon_i \quad for \ i = 1, \dots, n} \quad (1)$$

While this specification provides a useful point of departure for the study, it fails to identify the data generating process. The assumptions required for a linear-in-parameters model to be correctly identified are only likely to hold if conscientiousness is not affected by education, and conversely for conscientiousness to not be impacted by education. As was highlighted in Appendix ??, this assumption is violated through the bi-direction of correlation between variables. Thus, endogeneity tests are failed as a result of simultaneity between conscientiousness and education.

Furthermore, the construction of the conscientiousness variable is subject to measurement error. As conscientiousness is believed to be a latent measure, proxy questions for this skill were used through a principal components analysis to obtain conscientiousness scores for each individual. The imperfection of these questions as an exact measure of conscientiousness means that the error term in the conditional function is likely related to the conditional function itself, providing another source of endogeneity. As a result,  $\varepsilon_i$  contains useful information which is related to the

These issues prompt the need for a different identification method which takes them into account. Therefore, a structural modelling approach similar to that found in Heckman *et al.* (2006) was followed to allow simulation of a conscientiousness score from a specified structural form. This method solves both reverse causality and measurement error issues as it allows for all dependence across all measurements, choices and outcomes to come through the latent conscientiousness measure. Controlling for this dependence then allows for endogeneity in this model to also be controlled. For this method, it is assumed that the latent conscientiousness measure ( $f^c$ ) is important in generating both the responses to the observed proxy questions, as well as the labour market outcomes obtained. Education is also significant in this process as it may have affected the responses obtained to the questions at the times of the survey, and thus may

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<sup>2</sup>Factor loadings for each of the questions are found in Appendix ??.

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affect the intercept and factor loadings in the system. Conditioning on the latent conscientiousness score in a semiparametric structure then solves for the mentioned endogeneity concerns.

$$\boxed{Exercise_i(educ_i) = \beta_{0,1}(educ_i) + \alpha_1(educ_i)f_i^c + \varepsilon_{1,i}} \quad (2)$$

$$\boxed{PlanTert_i(educ_i) = \beta_{0,2}(educ_i) + \alpha_2(educ_i)f_i^c + \varepsilon_{2,i}} \quad (3)$$

$$\boxed{EmoHope_i(educ_i) = \beta_{0,3}(educ_i) + \alpha_3(educ_i)f_i^c + \varepsilon_{3,i}} \quad (4)$$

$$\boxed{Smoke_i(educ_i) = \beta_{0,4}(educ_i) + \alpha_4(educ_i)f_i^c + \varepsilon_{4,i}} \quad (5)$$

In the above equations,  $educ_i$  represents the education level at Wave 1 for individual  $i$ . Additionally the  $\beta$  and  $\alpha$  coefficients vary by education level to account for the fact that schooling affects measured conscientiousness. With the model for the conscientiousness factor now specified, it can now be entered as a regressor into the outcome equation for wages and employment status. Equation 6 illustrates this:

$$\boxed{\log(Wages_i) = \beta_0 + \alpha f_i^c + \gamma educ_i + \mathbf{X}'_i \boldsymbol{\delta} + \varepsilon_i \quad for \ i = 1, \dots, n} \quad (6)$$

Education was then modelled as an endogenous choice which depends on the latent conscientiousness measure. This controls for the fact that more conscientious individuals may select into higher levels of education.

$$\boxed{educ_i = \arg \max_j \{\beta_j \mathbf{X}_i + \alpha_j f_i^c + \varepsilon_{j,i}\} \quad for \ j \in \{\text{Education levels}\}} \quad (7)$$

The model is then identified, as per Heckman *et al.* (2006), when the following three assumptions are met. Firstly, a set of exclusion restrictions are implemented whereby family background variables affect educational choice but not labour outcomes directly. Next, to fix the scale of the latent conscientious measure  $\alpha_1$  is normalised = 1. Lastly, all dependence across the outcome variables occurs through  $f^c$  and the other observable measures.

With this the distributions of the scores obtained from the simulations are estimated nonparametrically through the use of Bayesian Monte Carlo Markov Chain methods. Posterior distributions of  $f^c$  for each individual are simulated. This allows for predictions of outcomes under different counterfactual conscientiousness levels to be calculated, whereafter the causal effect of moving individuals across the

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conscientiousness distribution can be obtained.

## 5. Results

This section outlines the results from both the linear estimation and structural modelling techniques. The former is flawed with endogeneity concerns, which sets the basis for the latter approach.

### 5.1. Linear Estimation

Conscientiousness plays a multitude of roles in the formation of various outcomes in an individual's life. Some of these outcomes include educational choice, behavioural decisions, wages, and others. To address issues of reverse causality, education and conscientiousness were modelled separately to determine if a causal relationship between the two variables was present.

Endogeneity issues arise through concerns of reverse causality between education and conscientiousness, as well as measurement error of the conscientiousness variable. As the table below demonstrates, conscientiousness has a significant positive role in the attainment of education. After the inclusion of control variables, this relationship remains significant. Appendix B lays out the result from the regression of education on conscientiousness, showing that the former is also positively correlated with the latter. This result strengthens the hypothesis of reverse causality between variables and reinforces the need for a method which accounts for this endogeneity introduced in the data generating process for the outcome variable. Allowing for this relationship to be accounted for through a simultaneous equation structural modelling approach, endogeneity issues can be reduced. Measurement error occurs as a result of the questions making up the variable of conscientiousness being imperfect proxies for the latter. By simulating factor scores of this measure through structural modelling simulation, more accurate scores are obtainable.

Table 1: Linear model results of conscientiousness on education

| Variables         | Dependent variable: Education |                      |
|-------------------|-------------------------------|----------------------|
|                   | (1)                           | (2)                  |
| Constant          | 12.062***<br>(0.052)          | 11.943***<br>(0.094) |
| Conscientiousness | 0.303***<br>(0.049)           | 0.367***<br>(0.060)  |
| Female            |                               | 0.695***<br>(0.126)  |
| Coloured          |                               | -0.755***<br>(0.177) |



|                |       |          |
|----------------|-------|----------|
| Asian/Indian   |       | 0.933    |
|                |       | (0.544)  |
| White          |       | 1.126*   |
|                |       | (0.498)  |
| Wealth         |       | 0.452*** |
|                |       | (0.042)  |
| N              | 5062  | 3516     |
| R-squared      | 0.007 | 0.066    |
| Adj. R-squared | 0.007 | 0.065    |

Note: \*\*\* p<0.001, \*\* p<0.01, \* p<0.05. Standard errors in parentheses.

To investigate the presence of a linear relationship between wages and conscientiousness, and employment status and conscientiousness a linear model was constructed. Results thereof are presented in table 2.

Table 2: Linear model results of conscientiousness on employment status and wages

| Variables              | Dependent variable: Employment status = 1 |                      |                      |                      |                      | Dependent variable: Log(Wages) |                      |                      |                      |                      |
|------------------------|-------------------------------------------|----------------------|----------------------|----------------------|----------------------|--------------------------------|----------------------|----------------------|----------------------|----------------------|
|                        | (1)                                       | (2)                  | (3)                  | (4)                  | (5)                  | (6)                            | (7)                  | (8)                  | (9)                  | (10)                 |
| Constant               | 0.474***<br>(0.008)                       | -2.300***<br>(0.339) | -2.338***<br>(0.331) | -1.910***<br>(0.382) | -1.910***<br>(0.382) | 7.422***<br>(0.019)            | 1.956*<br>(0.895)    | 2.162*<br>(0.886)    | 3.330**<br>(1.025)   | 3.330**<br>(1.025)   |
| Conscientiousness      | 0.045***<br>(0.007)                       | 0.035***<br>(0.007)  | -0.001<br>(0.007)    | 0.004<br>(0.008)     | 0.004<br>(0.008)     | 0.156***<br>(0.018)            | 0.120***<br>(0.017)  | 0.017<br>(0.018)     | 0.022<br>(0.021)     | 0.022<br>(0.021)     |
| Education              |                                           | 0.027*<br>(0.011)    | 0.043***<br>(0.011)  | 0.048***<br>(0.013)  | 0.048***<br>(0.013)  |                                | 0.102***<br>(0.026)  | 0.123***<br>(0.027)  | 0.077*<br>(0.033)    | 0.077*<br>(0.033)    |
| Education <sup>2</sup> |                                           | -0.000<br>(0.000)    | -0.001*<br>(0.000)   | -0.001*<br>(0.000)   | -0.001*<br>(0.000)   |                                | -0.001<br>(0.001)    | -0.001<br>(0.001)    | 0.000<br>(0.001)     | 0.000<br>(0.001)     |
| Age                    |                                           | 0.240***<br>(0.037)  | 0.238***<br>(0.036)  | 0.189***<br>(0.042)  | 0.189***<br>(0.042)  |                                | 0.409***<br>(0.096)  | 0.390***<br>(0.095)  | 0.302**<br>(0.110)   | 0.302**<br>(0.110)   |
| Age <sup>2</sup>       |                                           | -0.005***<br>(0.001) | -0.005***<br>(0.001) | -0.004***<br>(0.001) | -0.004***<br>(0.001) |                                | -0.009***<br>(0.003) | -0.009***<br>(0.003) | -0.006*<br>(0.003)   | -0.006*<br>(0.003)   |
| Female                 |                                           |                      | -0.210***<br>(0.015) | -0.200***<br>(0.017) | -0.200***<br>(0.017) |                                |                      | -0.578***<br>(0.039) | -0.573***<br>(0.045) | -0.573***<br>(0.045) |
| Coloured               |                                           |                      | 0.192***<br>(0.022)  | 0.208***<br>(0.024)  | 0.208***<br>(0.024)  |                                |                      | 0.295***<br>(0.054)  | 0.255***<br>(0.059)  | 0.255***<br>(0.059)  |
| Asian/Indian           |                                           |                      | 0.009<br>(0.072)     | 0.121<br>(0.079)     | 0.121<br>(0.079)     |                                |                      | 0.595**<br>(0.211)   | 0.719**<br>(0.227)   | 0.719**<br>(0.227)   |
| White                  |                                           |                      | 0.339***<br>(0.056)  | 0.348***<br>(0.071)  | 0.348***<br>(0.071)  |                                |                      | 1.064***<br>(0.170)  | 0.697***<br>(0.199)  | 0.697***<br>(0.199)  |
| Wealth                 |                                           |                      |                      | 0.002<br>(0.006)     | 0.002<br>(0.006)     |                                |                      |                      | 0.087***<br>(0.016)  | 0.087***<br>(0.016)  |
| N                      | 4269                                      | 4260                 | 4260                 | 3238                 | 3238                 | 3284                           | 3276                 | 3099                 | 2308                 | 2308                 |
| R-squared              | 0.009                                     | 0.086                | 0.142                | 0.135                | 0.135                | 0.023                          | 0.136                | 0.219                | 0.235                | 0.235                |
| Adj. R-squared         | 0.009                                     | 0.084                | 0.140                | 0.133                | 0.133                | 0.022                          | 0.134                | 0.217                | 0.231                | 0.231                |

Note: \*\*\* p<0.001, \*\* p<0.01, \* p<0.05. Standard errors in parentheses (robust standard errors reported in columns (1)-(5)).

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Simple models for conscientiousness and outcomes (columns (1) and (6)) illustrate a statistically significant positive effect of conscientiousness on both the probability of being employed and the log of wages earned. Including relevant control variables leads to a decrease in the magnitude of effect and loss of statistical significance. This result for the effect of conscientiousness on the outcomes can either mean that conscientiousness is not an important factor in determining employment status or wages (i.e. the error term in the simple model captured most of the information), or alternatively that in this linear setup the effects of conscientiousness are hidden due to endogeneity concerns as was previously mentioned. The latter claim motivates the extension of the analysis into a structural model to account for such concerns.

### *5.2. Structural Modelling Approach*

## **6. Discussion**

## **7. Conclusion**

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## Appendix

### *Appendix A: Conscientiousness PCA loadings*

| Variables                   | PC1    |
|-----------------------------|--------|
| Exercise                    | -0.682 |
| Emotional hope              | -0.630 |
| Planning tertiary education | -0.122 |
| Smoked                      | 0.351  |

*Note:* Obtained from PCA results

### *Appendix B: Regressing education on conscientiousness*

| Variables      | Dependent variable: Conscientiousness |                      |
|----------------|---------------------------------------|----------------------|
|                | (1)                                   | (2)                  |
| Constant       | -0.299***<br>(0.050)                  | -0.074<br>(0.062)    |
| Education      | 0.024***<br>(0.004)                   | 0.029***<br>(0.005)  |
| Female         |                                       | -0.610***<br>(0.034) |
| Coloured       |                                       | 0.258***<br>(0.049)  |
| Asian/Indian   |                                       | 0.155<br>(0.152)     |
| White          |                                       | 0.625***<br>(0.139)  |
| Wealth         |                                       | 0.026*<br>(0.012)    |
| N              | 5062                                  | 3516                 |
| R-squared      | 0.007                                 | 0.109                |
| Adj. R-squared | 0.007                                 | 0.107                |

*Note:* \*\*\* p<0.001, \*\* p<0.01, \* p<0.05. Standard errors in parentheses.